Band-1 Hadamard Order 6484 — Miyamoto (Paley-I H_1620 + C_1622 over prime 1621); Miyamoto-only Bonus (NOT in SPK); Gap 2052 Still OPEN

H_1620 Paley-I/II PASS; C_1622 PASS; H_6484 Miyamoto PASS; Paley-I/II on 6484 SKIP; 6484 Miyamoto-only (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_6436_Miyamoto`; `2026-09-15_Band1_Hadamard_6404_Kronecker`; `2026-09-15_Band1_Hadamard_6388_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_6484_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1620 and 6484 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with preferred Paley-I seed H_{1620} = Paley-I($q = 1619$) (also verified Paley-II seed $q = 809$) and symmetric Paley conference C_{1622} over prime $q = 1621 \equiv 1 \pmod 4$. Double half=810 **FAIL**s ($q = 809 \equiv 1 \pmod 4$). Classical SPK paths for order 6484 **FAIL**: Paley-I ($q = 6483 = 3 \cdot 2161$ not prime); Paley-II ($q = 3241 = 7 \cdot 463$ not prime); Kronecker (4×1621). Therefore order **6484 is a Miyamoto-only bonus (NOT in SPK)** (same honesty pattern as 6404/6388/6196/5956). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. `clay_closed = false`. Skip $q = 1613 \to n = 6452$ (no easy seed). Prior SPK seeds **1608** / **6436** (via Paley-II) retained; Miyamoto-only bonuses **6388/6404/6196/...** excluded.

Abstract

Finite constructions of orders **1620** and **6484** via Miyamoto.

- H_{1620} = Paley-I ($q = 1619$) — preferred; Paley-II ($q = 809$) also PASS; Double SKIP
- C_{1622} = Paley conference over prime 1621
- H_{6484} = Miyamoto(H_{1620}, C_{1622})
- Paley-I / Paley-II for 6484: **SKIP**

SPK: 6484 Miyamoto-only bonus (NOT in SPK). Seed 1620 enters SPK via Paley-I (and Paley-II). Prior SPK **1608** / **6436** retained in ONBOX.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_6484.py`

Results. overall_pass=true; units 82/82; wall 224.6319s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 6484 = 4 \cdot 1621$; 1621 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1620} and C_{1622} (Paley conference over prime 1621).
- Seed preferred: Paley-I $q = 1619$ (prime $\equiv 3 \pmod 4$); $q + 1 = 1620$.
- Also eligible: Paley-II $q = 809$ (prime $\equiv 1 \pmod 4$); $2(q + 1) = 1620$.
- Double half=810 **FAIL** ($809 \equiv 1 \pmod 4$).
- SPK: Paley-I $q = 6483$ **FAIL**; Paley-II $q = 3241$ **FAIL**; Kronecker **FAIL** → **Miyamoto-only**.
- Skip: $q = 1613 \rightarrow n = 6452$ (no easy seed).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_1620 Paley-I	Paley-I q=1619 (preferred)	0	3.2156
H_1620 Paley-II	Paley-II q=809 (also eligible)	0	2.1972
H_1620 Double	SKIP q=809 1 mod 4	—	—
C_1622	Paley conf q=1621	0	3.0235
H_6484 Miyamoto	Miyamoto(H_1620 Paley-I, C_1622)	0	213.0507
H_6484 Paley-I	SKIP q=6483=3 · 2161 not prime	—	—
H_6484 Paley-II	SKIP q=3241=7 · 463 not prime	—	—

SPK closure note

Seeds include prior SPK ships through **6436** lineage (**1608** Paley-I; **6436** Paley-II; **1600** Kronecker chain; **1596** Paley-II; **NOT** 6388/6404/6196) + verified **1620** (Paley-I / Paley-II); **6484 NOT folded** (Miyamoto-only); **exclude** Miyamoto-only bonuses including **6484/6404/6388/6196/5956/5812/5476**. Known count 1329. Next constructive gap **2052** (OPEN).

Quantity	Value
6484 in SPK	**False** (Miyamoto-only bonus)
1620 in SPK	**True** (Paley-I / Paley-II)
6436 in SPK	**True** (prior Paley-II)
1608 in SPK	**True** (prior Paley-I)
6404 / 6388 in SPK	**False** (prior Miyamoto-only)
Next constructive gap	**2052** (OPEN)

Next Miyamoto targets (after q=1621; skipped 1613→6452)

q	n=4q	need H	build notes
1637	6548	1636	[] (no easy)
1657	6628	1656	[‘Double(PaleyI half=828)’]
1669	6676	1668	[‘PaleyI(q=1667)’]
1681	6724	1680	[‘Double(PaleyI half=840)’, ‘Kronecker(4x420)’]
1693	6772	1692	[]
1697	6788	1696	[‘Kronecker(4x424)’]

Honesty

clay_closed = false · GATE-04 LOCKED · Hadamard conjecture OPEN · gap 2052 OPEN · no RH/P!=NP/YM/Frankl/NS/AC/Clay-closed claims · 6484 Miyamoto-only (NOT in SPK).